

Phys5350 Assignment 2

1. For an unknown sample, preliminary XPS analysis showed a signal at the binding energy (BE) of about 532 eV. The signal at this BE can arise from either Pd 3p_{3/2} or O 1s. Suggest how you can confirm the presence of either Pd or O through further XPS experiments or data analysis.
2. How many XPS core level peaks do you expect for the element K? Count doublets as two and triplets as three, etc, but do not count small chemically shifted peaks as separate peaks.
3. Explain why H and He cannot be detected by Auger electron spectroscopy.